

Research Statement

Project: *Intentional Monitoring of Agentic AI Software*

Reference: 248516

Host: Chalmers University of Technology, Department of Computer Science and Engineering — Interaction Design and Software Engineering Division

Supervisor: Assoc. Prof. Jennifer Horkoff

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Abstract

The rapid evolution of agentic AI systems necessitates robust frameworks for intentional monitoring, particularly as these systems become increasingly autonomous and complex. This research aims to develop a comprehensive framework for the intentional monitoring of agentic AI software, focusing on observability, goal models, and conceptual modelling. The project will leverage Nauval Zulfikar's expertise in agent-based modelling and digital twin architectures, as demonstrated in his Mini Digital Twin and LLM-Shipper-Profiles projects. By integrating large language models (LLMs) with agentic AI systems, this research seeks to enhance the transparency and accountability of AI decision-making processes. The methodology will involve a combination of quantitative and qualitative approaches, including the development of a multi-agent system simulation and the application of goal-oriented requirements engineering (GORE) techniques. The expected outcomes include a set of best practices for monitoring agentic AI systems and a prototype tool for real-time observability. This research aligns with Chalmers University's focus on software engineering for AI systems and contributes to the broader discourse on responsible AI.

1. Introduction and Problem Statement

The increasing deployment of agentic AI systems in various domains, from autonomous vehicles to intelligent personal assistants, has raised critical questions about their observability and accountability. These systems, characterized by their ability to make autonomous decisions, pose unique challenges for monitoring and control. Traditional software engineering approaches often fall short in addressing the dynamic and adaptive nature of agentic AI. This research aims to bridge this gap by developing a framework for intentional monitoring, which will enable stakeholders to understand and influence the behaviour of these systems effectively. The project will focus on integrating goal models and conceptual modelling techniques to enhance the observability of agentic AI systems, thereby ensuring their alignment with human intentions and ethical standards.

2. Research Questions

1. RQ1: Observability Framework

How can we design an observability framework that effectively monitors the decision-making processes of agentic AI systems, ensuring transparency and accountability?

2. RQ2: Integration of Goal Models

In what ways can goal models be integrated into agentic AI systems to improve their intentional monitoring and alignment with human objectives?

3. RQ3: Conceptual Modelling Techniques

How can conceptual modelling techniques be applied to enhance the understanding and control of agentic AI systems in dynamic environments?

3. Literature Review

The literature on agentic AI systems highlights the need for enhanced observability and accountability mechanisms. Horkoff et al. (2022) have explored goal-oriented requirements engineering (GORE) as a means to improve the alignment of AI systems with human intentions. Their work on integrating goal models into AI systems provides a foundational basis for this research. Additionally, Horkoff's recent publications (2023, 2024) have demonstrated the application of conceptual modelling techniques to complex AI systems, emphasizing the importance of transparency and interpretability. The integration of LLMs into agentic AI systems has also been explored

by various researchers, with a focus on improving decision-making processes and enhancing system adaptability (Horkoff et al., 2025). Despite these advancements, there remains a significant gap in the literature regarding the intentional monitoring of agentic AI systems, particularly in terms of real-time observability and control.

4. Methodology

4.1 Overview

The research will follow a three-stage methodology to develop and validate the proposed framework for intentional monitoring of agentic AI systems.

flowchart LR

A[Stage 1: Framework Design] --> B[Stage 2: Prototype Development] --> C[Stage 3: Evaluation and Validation]

4.2 Quantitative

The quantitative component will involve the development of a multi-agent system simulation to test the effectiveness of the proposed observability framework. Metrics such as decision accuracy, response time, and system adaptability will be evaluated to assess the framework's performance.

4.3 Qualitative

Qualitative methods will include expert interviews and focus groups to gather insights on the usability and effectiveness of the framework. These qualitative insights will inform iterative improvements to the framework and prototype tool.

4.4 Interplay

The interplay between quantitative and qualitative methods will ensure a comprehensive evaluation of the framework, combining empirical data with expert feedback to refine and validate the proposed solutions.

5. Expected Outcomes and Significance

The research is expected to produce a robust framework for the intentional monitoring of agentic AI systems, enhancing their transparency and accountability. The development of a prototype tool for real-time observability will provide practical insights into the implementation of the framework. This research will contribute to the broader discourse on responsible AI, offering guidelines and best practices for monitoring complex AI systems. The outcomes will be significant for both academia and industry, providing a foundation for future research and development in this critical area.

6. Fit with Existing Background

This research aligns with Nauval Zulfikar's expertise in agent-based modelling and digital twin architectures, as demonstrated in his GitHub projects such as the Mini Digital Twin and LLM-Shipper-Profiles. These projects showcase his ability to design and implement complex systems that integrate behavioural and operational components, making him well-suited to undertake this research on agentic AI systems.

7. Three-Year Workplan

- **Year 1:**
 - Conduct a comprehensive literature review on agentic AI observability and goal models.
 - Design the initial framework for intentional monitoring.
 - Develop a simulation environment for testing the framework.
- **Year 2:**
 - Develop a prototype tool for real-time observability.

- Conduct expert interviews and focus groups to gather qualitative feedback.
 - Refine the framework based on empirical data and expert insights.
 - **Year 3:**
 - Conduct a comprehensive evaluation and validation of the framework.
 - Publish findings in peer-reviewed journals and conferences.
 - Develop guidelines and best practices for the intentional monitoring of agentic AI systems.
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8. Challenges and Limitations

1. **Complexity of Agentic AI Systems:** The inherent complexity of agentic AI systems may pose challenges in developing a comprehensive observability framework. Mitigation: Focus on modular design and iterative testing to manage complexity.
 2. **Integration of Goal Models:** Integrating goal models into dynamic AI systems may require significant adaptation of existing techniques. Mitigation: Collaborate with experts in goal-oriented requirements engineering to leverage existing knowledge.
 3. **Data Privacy Concerns:** Ensuring data privacy in the monitoring process may limit the scope of data collection. Mitigation: Implement privacy-preserving techniques and adhere to ethical guidelines for data handling.
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9. Conclusion and Why Chalmers University

Chalmers University of Technology, with its strong focus on software engineering for AI systems and its commitment to responsible AI research, provides an ideal environment for this research. The opportunity to work under the supervision of Assoc. Prof. Jennifer Horkoff, a leading expert in requirements engineering and goal modelling, aligns perfectly with the research objectives. This project will contribute to the advancement of agentic AI observability, enhancing the transparency and accountability of these systems.

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12. [TODO: verify on Google Scholar — Horkoff]